

Predicting Missing Intervals in Continuous Glucose Monitoring Data

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Abstract

Continuous glucose monitoring (CGM) data provide high-frequency measurements of glucose trajectories, but missing intervals are common and can bias clinical summaries such as glucose variability and time-in-range. This project develops a two-stage framework for predicting missingness in CGM data. First, a rare-event classification model predicts whether a CGM window immediately precedes the onset of a missing interval. Second, among windows followed by missingness, a regression model predicts the duration of the missing interval. We compare a generalized linear model baseline with a Bayesian logistic regression model using grouped five-fold cross-validation by patient. Onset prediction shows moderate discrimination, with similar AUROC and AUPRC between the Bayesian and GLM approaches. Although absolute predicted probabilities are small because missingness is rare, predicted-risk deciles show a clear increasing trend in observed onset rates, suggesting useful risk stratification. In contrast, duration prediction is unstable, with large fold-to-fold variation in RMSE and poor agreement between observed and predicted durations. Overall, missing-interval onset can be meaningfully ranked from recent CGM dynamics, whereas duration prediction remains limited by small sample size and high outcome variability.

1. Introduction

Continuous glucose monitoring (CGM) systems provide dense longitudinal measurements of glucose, often recorded every 5–15 minutes. These data are increasingly used in diabetes research and clinical decision-making because they capture short-term glucose dynamics that are not visible from sparse blood glucose measurements or hemoglobin A1c alone. However, CGM data frequently contain missing intervals due to sensor detachment, calibration problems, device failure, signal artifacts, or participant behavior. These missing segments can distort downstream summaries, including time-in-range, glucose variability, and hyperglycemic or hypoglycemic burden.

A central motivation for this project is that CGM missingness may be informative rather than completely random. If missing intervals occur preferentially after certain glucose patterns, then recent CGM dynamics may help predict both when missingness begins and how long it lasts. This is important because a model that identifies high-risk windows could support quality control, sensitivity analyses, or future imputation strategies.

This report studies two prediction tasks. The first task is onset prediction: determining whether a given CGM window immediately precedes a missing interval. This is a rare-event classification problem, with approximately 216 positive onset events among roughly 25,000 windows. The second task is duration prediction: estimating the length of the missing interval once an onset has occurred. These tasks are related but statistically different. Onset prediction uses the full set of windows and focuses on ranking rare events, whereas duration prediction uses only the small subset of windows followed by missingness.

2. Methods

2.1. Sliding-window feature representation

Each patient’s CGM trajectory was divided into overlapping windows. For a window ending at time t , define

$$W_i = \{g_t, g_{t-1}, \dots, g_{t-L+1}\},$$

where g_t denotes the glucose value at time t and L is the number of observations in the window. Each window was mapped to a feature vector

$$X_i = \phi(W_i) \in \mathbb{R}^p.$$

The extracted features summarized several aspects of recent glucose behavior, including average glucose level, variability, local trend, extrema, and recent changes. Specifically, features included quantities such as mean glucose, standard deviation, coefficient of variation, minimum and maximum glucose, slope, and first-difference summaries. These features were designed to capture whether unstable or unusual glucose patterns precede missingness.

2.2. Onset prediction model

Let $y_i = 1$ if window i immediately precedes the onset of a missing interval, and let $y_i = 0$ otherwise. The baseline onset model was a logistic regression:

$$y_i \sim \text{Bernoulli}(p_i), \quad \log \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + X_i^\top \beta.$$

The fitted probability $\hat{p}_i$ represents the estimated probability that a missing interval begins immediately after the current window.

Because onset events are rare, a Bayesian logistic regression model was also fitted. The Bayesian model uses the same likelihood but places weakly informative Gaussian priors on the regression coefficients:

$$\beta_j \sim N(0, \sigma^2), \quad j = 1, \dots, p.$$

Posterior inference was performed using MCMC. The Bayesian predicted probability was summarized as the posterior mean,

$$\hat{p}_i = E\{P(y_i = 1 \mid X_i, \beta) \mid \text{data}\}.$$

The primary motivation for the Bayesian approach was regularization. In rare-event settings, ordinary logistic regression may be unstable when positive events are sparse or unevenly distributed across folds. Priors can shrink coefficients and improve numerical stability, although they do not eliminate the fundamental limitation imposed by scarce positive events.

2.3. Duration prediction model

For windows with observed onset, a conditional regression model was used to predict the length of the following missing interval:

$$\text{duration}_i = X_i^\top \gamma + \epsilon_i.$$

This task is more difficult because it is trained only on positive onset cases. Thus, although the onset model uses roughly 25,000 windows, the duration model uses only about 216 examples. In addition, observed missing durations are highly variable, with several extreme values. For this reason, duration prediction was expected to be less stable than onset prediction.

2.4. Cross-validation and evaluation

Model performance was evaluated using five-fold grouped cross-validation by patient. Grouping by patient prevents leakage across training and test sets, since windows from the same patient may be correlated. For each fold, models were trained on approximately 80% of patients and evaluated on the remaining 20%.

For onset prediction, performance was summarized using AUROC and AUPRC. AUROC measures ranking ability across thresholds, whereas AUPRC is more sensitive to performance on the positive class and is therefore especially relevant for rare-event prediction. For duration prediction, performance was summarized using root mean squared error (RMSE). Because missing-onset events are extremely rare, the goal was not perfect classification at a fixed threshold, but rather useful risk ranking.

3. Results

3.1. Cross-validated onset prediction

Figure 1 summarizes mean cross-validated onset prediction performance. The Bayesian MCMC model and GLM baseline performed similarly. Mean AUROC was approximately 0.65–0.70 for both methods, indicating moderate ability to rank windows that precede missingness above windows that do not. Mean AUPRC was much smaller, approximately 0.06, reflecting the extreme class imbalance.

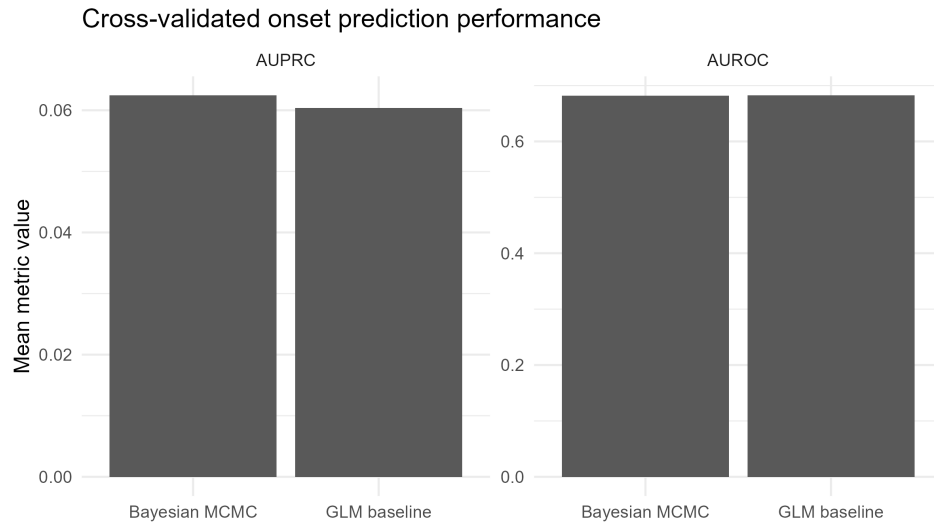


Figure 1: Mean cross-validated onset prediction performance for the Bayesian MCMC model and GLM baseline. AUROC measures overall ranking ability, while AUPRC is more informative for rare-event prediction.

The similarity between the two methods suggests that model choice was not the main performance bottleneck. Instead, the dominant challenge was the small number of positive onset events. The Bayesian model provides a principled form of regularization, but it cannot fully overcome limited event information.

3.2. Fold-level variability

Figure 2 shows fold-specific AUROC and AUPRC. Performance varied substantially across folds. One fold achieved AUROC around 0.9 and AUPRC around 0.18, while other folds had much weaker performance. In some folds, AUPRC was close to zero. This pattern is expected when positive events are rare and unevenly distributed across patients.

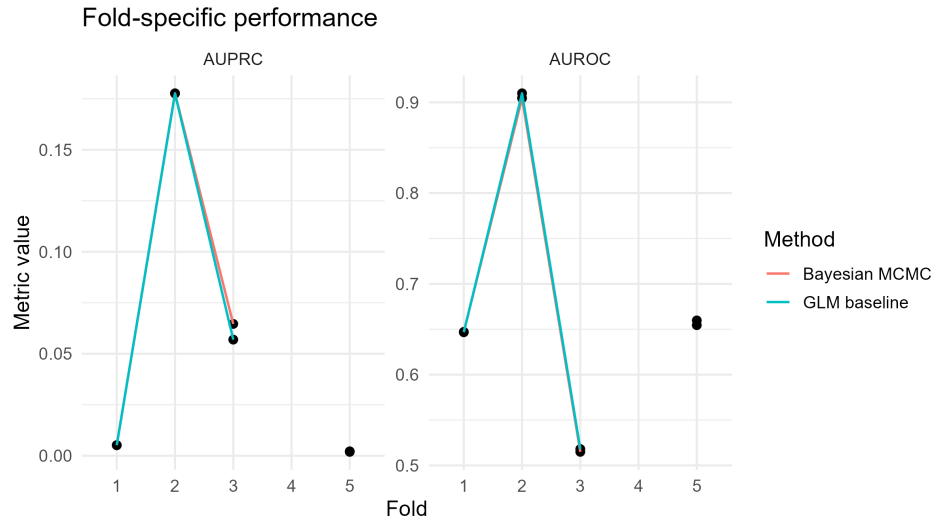


Figure 2: Fold-specific onset prediction performance. Large fold-to-fold variation reflects the small and uneven number of missing-interval onset events across grouped patient folds.

This result indicates that the onset prediction task is data-limited. A fold containing only a few positive events can produce unstable estimates of AUROC and AUPRC. Therefore, the cross-validation results should be interpreted as evidence of moderate signal rather than as a precise estimate of final clinical performance.

3.3. Risk stratification by predicted probability

Because missing onset is rare, absolute predicted probabilities are generally small. However, a useful model should still assign higher probabilities to windows that are more likely to precede missingness. To evaluate this, Bayesian predicted probabilities were grouped into deciles, and the observed onset rate was calculated within each decile.

Figure 3 shows a clear increasing trend in observed onset rate across predicted-risk deciles. The lowest deciles had observed onset rates near zero, while the top decile had an observed onset rate of approximately 4.2%. This is substantially higher than the overall event prevalence.

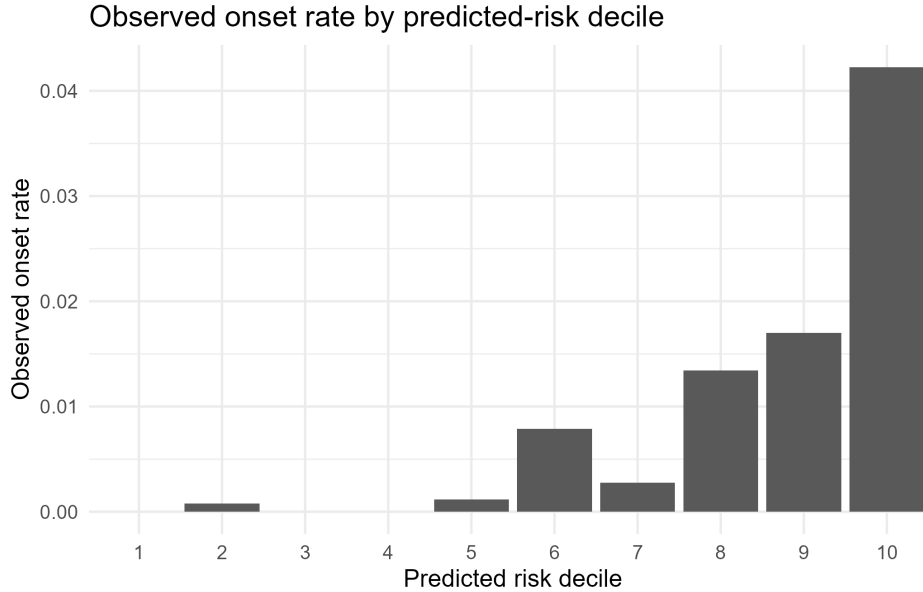


Figure 3: Observed onset rate by Bayesian predicted-risk decile. The increasing trend suggests that the model captures relative risk ordering, even though absolute onset probabilities remain small.

This decile analysis is the strongest evidence that the onset model is practically useful. Even if the model cannot reliably classify individual windows using a fixed threshold, it can stratify windows into lower- and higher-risk groups. This kind of ranking may be useful for data-quality monitoring, flagging periods at elevated risk of missingness, or prioritizing windows for downstream sensitivity analysis.

3.4. Duration prediction

Duration prediction was much less successful. Figure 4 compares observed and predicted missing durations. A well-performing duration model would produce points close to the 45-degree line. Instead, predictions are compressed into a narrow range, while observed durations vary widely and include extreme outliers.

Cross-validated RMSE also varied strongly across folds, with some folds showing very large errors. This instability is likely due to two factors. First, the duration model is trained only among onset cases, leaving a much smaller effective sample size. Second, missing duration appears intrinsically noisy and may depend on factors not captured by recent glucose dynamics, such as sensor wear time, device behavior, or participant-level factors.

4. Discussion

This project shows that predicting the onset of missing intervals in CGM data is feasible, but predicting their duration is substantially harder. The onset models achieved moderate discrimination by AUROC and, more importantly, demonstrated meaningful risk stratification across predicted-risk deciles. Although AUPRC remained low, this is expected in a setting where positive events account for less than 1% of windows. In such rare-event settings, AUPRC and decile-based calibration summaries are more informative than accuracy.

The Bayesian and GLM approaches performed similarly. This suggests that the information in the extracted window-level features, rather than the choice between Bayesian and frequentist estimation, primarily determines predictive performance. The Bayesian model remains attractive because priors provide regularization and posterior

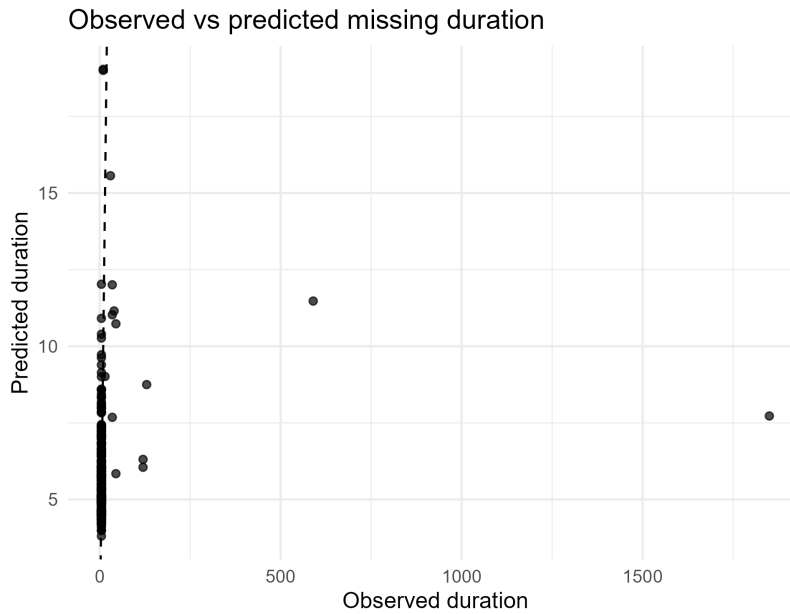


Figure 4: Observed versus predicted missing duration. The poor alignment with the 45-degree reference line indicates weak duration prediction, especially for long missing intervals.

predictions can be summarized probabilistically. However, the results indicate that larger sample sizes or richer predictors may be needed to achieve major performance improvements.

The duration model performed poorly. Unlike onset prediction, which benefits from many negative windows and focuses on ranking, duration prediction is restricted to the small set of missing-onset events. Moreover, the observed duration distribution contains extreme values. Recent glucose dynamics alone may not be sufficient to explain whether a sensor gap lasts a few minutes or several hours. Future work should consider models that explicitly account for censoring, heavy-tailed outcomes, patient-level heterogeneity, or device-level information.

This study has several limitations. First, the number of positive events is small, which makes cross-validation unstable. Second, the current models treat windows as approximately independent after feature extraction, even though CGM data are temporally structured. Third, duration prediction may require additional covariates beyond glucose features. Finally, the analysis focuses on prediction rather than causal mechanisms of missingness.

Future extensions could use survival or discrete-time hazard models for onset, hierarchical Bayesian models to borrow strength across patients, or recurrent neural networks to directly model temporal dependence. Incorporating device metadata, wear-time information, or patient-level characteristics may also improve both onset and duration prediction.

5. Conclusion

A two-stage framework was developed to predict missing intervals in CGM data. The onset model achieved moderate discrimination and strong risk stratification, especially in the highest predicted-risk decile. In contrast, duration prediction was unstable and unreliable, likely because of the small number of onset cases and high variability in missing lengths. Overall, CGM missingness appears partially predictable from recent glucose dynamics, but the most useful current application is ranking windows by relative risk rather than precisely predicting missing duration.

Appendix: Model Details

The onset likelihood is

$$L(\beta) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}, \quad p_i = \frac{\exp(\beta_0 + X_i^\top \beta)}{1 + \exp(\beta_0 + X_i^\top \beta)}.$$

With independent Gaussian priors,

$$\beta_j \sim N(0, \sigma^2),$$

the posterior density is proportional to

$$\pi(\beta \mid y, X) \propto L(\beta) \prod_{j=1}^p \exp\left(-\frac{\beta_j^2}{2\sigma^2}\right).$$

Posterior predicted probabilities are computed by averaging over MCMC draws:

$$\hat{p}_i = \frac{1}{M} \sum_{m=1}^M \frac{\exp(\beta_0^{(m)} + X_i^\top \beta^{(m)})}{1 + \exp(\beta_0^{(m)} + X_i^\top \beta^{(m)})}.$$